

Sony TA-F3A integrated stereo amplifier (1977) – documentation report

Introduction

Sony's TA-F3A is an integrated stereo amplifier introduced in the late 1970s. Sony's support site acknowledges the model but provides no manuals ¹, so our documentation must come from service manuals and independent sources. The amplifier was a minor update to the TA-F3 and combined power amplification, pre-amplification and phono equalisation in one unit. It shared many design features with the TA-F3 while increasing input bandwidth and changing the power supply and dimensions. The table below summarises the key specifications and features gathered from the service manual and contemporary literature.

Key specifications

Parameter	TA-F3A specification	Evidence
Power output (continuous RMS)	50 W + 50 W (8 Ω both channels driven) at 1 kHz and across 20 Hz–20 kHz with $\leq 0.5\%$ total harmonic distortion (THD) ² ³ . DIN 45500 rating (AEP/UK/E) 52 W + 52 W ² .	Service manual's specifications page ² .
Power bandwidth (IHF)	10 Hz–25 kHz ² .	Manual ² ; the Marktplaats listing notes a wider frequency response of 10 Hz–40 kHz ⁴ , which may be marketing rather than IHF bandwidth.
Frequency response	For the TA-F3A the frequency response is 10 Hz–60 kHz (± 1 dB) according to Hi-Fi Engine ⁵ . Sellers often state 10 Hz–40 kHz ⁴ .	Hi-Fi Engine spec sheet ⁵ and Marktplaats listing ⁴ .
Total harmonic distortion (THD)	$\leq 0.5\%$ at rated output; $\leq 0.1\%$ at 1 W ² . Later sources claim THD as low as 0.08 % for the TA-F3 ⁶ but the TA-F3A manual emphasises $\leq 0.5\%$ ² .	Manual ² ; Audio-database article for TA-F3 ⁶ .
Damping factor	30 ² (TA-F3: 40 ⁶).	Manual ² .

Parameter	TA-F3A specification	Evidence
Input sensitivity / impedance	MM phono: ~2.5 mV / 47 k Ω ; Line (tape, tuner, auxiliary): 150 mV (AEP/E models) or 200 mV (F3A as noted by Hi-Fi Engine) ⁵ . Impedance 47 k Ω for phono and 50 k Ω for line inputs ⁵ .	Hi-Fi Engine specification table ⁵ .
Signal-to-noise ratio (SNR)	75 dB for MM phono; 90 dB for line inputs ⁴ ⁷ .	Marktplaats and Atherstone lists ⁴ ⁷ .
Speaker load impedance	8 Ω or 16 Ω ⁴ .	Marktplaats/Atherstone specifications ⁴ .
Power requirements / consumption	US/Canadian model: 120 V AC, 60 Hz, 85 W; AEP model: 220 V AC, 50/60 Hz, 210 W; UK model: 240 V AC, 50/60 Hz, 240 W; E model: 120/220 V AC switchable, 50/60 Hz, 210 W ² .	Manual's general specifications ² .
Dimensions	US/Canadian model: 435 mm $\times$ 145 mm $\times$ 370 mm; AEP/UK/E models: 410 mm $\times$ 145 mm $\times$ 370 mm ² .	Manual ² .
Weight	~10.5 kg (US/Canada) or 9.5 kg (AEP/UK/E) net; shipping weight ~12.5 kg or 11.5 kg ² . Marketplace listings cite 11.1 kg ⁴ ⁷ .	Manual ² ; Marktplaats and Atherstone adverts ⁴ ⁷ .
Semiconductors	34 transistors and 16 diodes ⁵ .	Hi-Fi Engine specification table ⁵ .

Notes: other sources list the TA-F3A's frequency response as 10 Hz–40 kHz or 10 Hz–60 kHz, and THD at 0.08 %. These discrepancies arise because sellers sometimes combine F3A data with the earlier TA-F3's more ambitious specifications ⁶.

Design and circuit features

Power amplifier architecture

Like the earlier TA-F3, the TA-F3A employed a “pure-complementary, all-stage direct-coupled” power amplifier. A dual-transistor differential amplifier feeds a complementary push–pull output stage using bipolar transistors. To reduce inter-channel crosstalk, the pre-amp has a **FET buffer power supply**—a field-effect transistor follower that isolates the voltage amplifier from the power supply noise ⁶. The amplifier also uses **direct coupling** (no coupling capacitors) throughout the audio path for improved phase response ⁶.

Control circuitry

Sony emphasised tone neutrality and included a **self-defeat tone control** that disengages the tone circuit when controls are set to the 0 position, leaving a flat frequency response ⁶. A **muting circuit** delays the speaker outputs when switching on or off, preventing “pops,” and the service manual describes a protection circuit that detects output DC and shuts down the relay for safety ². A **power meter** (0.01 W–100 W) on the front panel displays output levels; it can be calibrated to show dynamic peaks and assists in matching amplifier power to speakers ⁶. Filter and loudness controls provide low-frequency boost at low volume and high-frequency attenuation to reduce hiss ⁶.

Inputs and connectors

The TA-F3A provides inputs for phono (moving-magnet), tuner, auxiliary and two tape decks (tape 1/2). Input sensitivity is around 2.5 mV for phono and 150–200 mV for line, with input impedance of 50 k Ω ⁵. Both **RCA** and **DIN** connectors were supplied, catering to European and Japanese audio standards ⁴. Users could connect two pairs of speakers, but the amplifier’s speaker selector allowed only A or B, not A+B simultaneously, to maintain proper impedance.

Dimensions and construction

The TA-F3A’s physical design emphasised sturdy construction. US/Canadian models are wider (435 mm) than AEP/UK/E models (410 mm) ². Advertisements describe a heavy steel case and a large power transformer, weighing roughly 10.5 kg ². Market listings mention that the phono stage uses a large discrete RIAA equaliser board that gives a **warm, full sound** ⁸. The front panel includes aluminium knobs for volume, balance, bass, treble, loudness and muting, along with a headphone jack and two sets of speaker terminals.

User experiences and reliability

Positive impressions

Vintage hi-fi enthusiasts value the TA-F3A for its **warm phono stage**, dynamic power and robust build quality. Sellers on Reverb describe restored units as sounding “warm” with an “effective loudness control” and note that after cleaning, replacing worn components and running for eight hours, the amplifier exhibited no issues ⁸. The self-defeat tone control and muting circuits are praised for delivering a neutral sound and preventing pops. ⁸ ⁹ The combination of FET input stage and discrete output stage yields a smooth sound with sufficient headroom for 8 Ω or 16 Ω speakers ⁴.

Caveats and service notes

An Audiokarma forum discussion notes that one unit was manufactured on **24 July 1978** and warns owners not to plug a **line-level source into the phono input**, because the phono equaliser can be damaged ¹⁰. The service manual includes a safety warning: shaded components in the schematic are safety-critical and must be replaced with Sony parts ². Many units require recap and cleaning due to age; sellers often replace electrolytic capacitors and adjust bias to factory specifications ⁸. The amplifier is not designed for 4 Ω loads, so running it with low-impedance speakers may cause overheating.

Differences from the TA-F3 (non-A)

The TA-F3A is often confused with the earlier TA-F3. The TA-F3 had a slightly wider frequency response (3 Hz–50 kHz), lower rated THD (0.08 %), higher damping factor (40), and weighed 8.7 kg ⁶. The TA-F3A replaced the F3's 3 Hz–50 kHz response with a 10 Hz–60 kHz bandwidth and reduced the damping factor to 30 ⁵. It also increased physical dimensions and weight. Because Sony's support site lists no manual for the TA-F3A ¹, many later sources merge F3 and F3A data, so it is important to differentiate them.

Conclusion

Although Sony's official site no longer provides a user manual for the TA-F3A ¹, service manuals and enthusiast resources reveal a well-engineered integrated amplifier. The TA-F3A delivers 50 W per channel into 8 Ω with a 10 Hz–60 kHz bandwidth and damping factor of 30 ² ⁵. Its design features—FET buffer power supply, complementary direct-coupled circuitry, self-defeat tone control and muting circuit—reflect Sony's attempt to combine high fidelity with user convenience ⁶. Vintage hi-fi enthusiasts appreciate its warm phono stage and robust construction ⁸, though they caution against overdriving the phono input ¹⁰. Overall, the TA-F3A stands as a notable late-1970s integrated amplifier that bridges the gap between classic discrete transistor designs and the more advanced integrated circuits of the 1980s.

¹ Manuals for TA-F3A | Sony USA

<https://www.sony.com/electronics/support/product/ta-f3a/manuals>

² ³ sony_ta-f3a.pdf_2.png (703×360)

https://elektrotanya.com/PREVIEW/63463243/23432455/sony/sony_ta-f3a.pdf_2.png

⁴ ≥ Sony TA-F3A versterker (1977-1979) — Versterkers en Receivers — Marktplaats

<https://www.marktplaats.nl/v/audio-tv-en-foto/versterkers-en-receivers/m2291368565-sony-ta-f3a-versterker-1977-1979>

⁵ Sony TA-F3 Integrated Stereo Amplifier Manual | HiFi Engine

https://www.hifiengine.com/manual_library/sony/ta-f3.shtml

⁶ SONY TA-F3 Specifications SONY

<https://audio-database.com/SONY-ESPRIT/amp/ta-f3.html>

⁷ Sony TA-F3A + ST-A3L (Sold) | My Site

<https://www.atherstone-audio.co.uk/product-page/sony-ta-f3a-st-a3l-amplifier-tuner>

⁸ ⁹ Sony TA-F3A Stereo Amplifier/ Total 100 Watts/ Fully Serviced & Tested / Excellent Condition / Free Shipping | Reverb

<https://reverb.com/item/69399571-sony-ta-f3a-stereo-amplifier-total-100-watts-fully-serviced-tested-excellent-condition-free-shipping>

¹⁰ Sony TA-F3A: "Restoring" Something I Shouldn't Have Bought | Audiokarma Home Audio Stereo Discussion Forums

<https://audiokarma.org/forums/index.php>